Markowitz2EPG

$$L(\vec{\omega}) = \vec{\omega}^T \Sigma \vec{\omega} - \lambda \mu^T \vec{\omega} \tag{1}$$

- $\vec{\omega}$: Vektor bobot aset.
- Σ: Matriks kovariansi dengan elemen σ_{ij}.
- μ: Matriks *expected return* dengan elemen μ_i.
- λ: Parameter *risk tolerance*.

$$\lambda = \frac{2}{\gamma} \implies \frac{1}{\lambda} = \frac{\gamma}{2} \tag{2}$$

- γ: Parameter *risk aversion*.

$$\vec{\omega} = \frac{\vec{x}}{k}, \quad k \in \mathbb{R} \tag{3}$$

- $\vec{x}$: Vektor strategi dalam *game*.
- k: Parameter skalar (mengingat $\sum_i \omega_i = \sum_i \frac{x_i}{k}$).

$$L(\vec{x}) = \left(\frac{\vec{x}}{k}\right)^T \Sigma \left(\frac{\vec{x}}{k}\right) - \lambda \mu^T \left(\frac{\vec{x}}{k}\right) \tag{4}$$

$$L(\vec{x}) = \frac{1}{k^2} \vec{x}^T \Sigma \vec{x} - \frac{\lambda}{k} \mu^T \vec{x} \tag{5}$$

$$\vec{x}^T \Sigma \vec{x} = \sum_{i=1}^{N} \sum_{j=1}^{N} \sigma_{ij} x_i x_j \tag{6}$$

$$\mu^T \vec{x} = \sum_{i=1}^{N} \mu_i x_i \tag{7}$$

$$U(\vec{x}) = L(\vec{x}) \cdot \left(-\frac{1}{\lambda}\right) \tag{8}$$

$$U(\vec{x}) = \left(\frac{1}{k^2} \sum_{i=1}^{N} \sum_{j=1}^{N} \sigma_{ij} x_i x_j - \frac{\lambda}{k} \sum_{i=1}^{N} \mu_i x_i\right) \cdot \left(-\frac{\gamma}{2}\right) \tag{9}$$

$$U(\vec{x}) = \frac{1}{k} \sum_{i=1}^{N} \mu_i x_i - \frac{\gamma}{2k^2} \sum_{i=1}^{N} \sum_{j=1}^{N} \sigma_{ij} x_i x_j \tag{10}$$

- $U(\vec{x})$: Fungsi potensial *Exact Potential Game*.

$$u_i(x_i, \vec{x}_{-i}) = U(x_i, \vec{x}_{-i}) - U(0, \vec{x}_{-i}) \tag{11}$$

$$u_i(x_i, \vec{x}_{-i}) = x_i \left(\frac{\mu_i}{k} - \frac{\gamma \sigma_{ii} x_i}{2k^2} - \frac{\gamma}{k^2} \sum_{j \neq i} \sigma_{ij} x_j \right) \tag{12}$$

Skenario A: $x_i = 0 \to x_i' = 1$

$$\Delta u_i(0 \to 1) = \frac{\mu_i}{k} - \frac{\gamma \sigma_{ii}}{2k^2} - \frac{\gamma}{k^2} \sum_{j \neq i} \sigma_{ij} x_j \tag{13}$$

Skenario B: $x_i = 1 \to x_i' = 0$

$$\Delta u_i(1 \to 0) = -\frac{\mu_i}{k} + \frac{\gamma \sigma_{ii}}{2k^2} + \frac{\gamma}{k^2} \sum_{j \neq i} \sigma_{ij} x_j \tag{14}$$

- Δu_i: Perubahan utilitas marginal pemain i.

Kasus Khusus ($k = 1$):

$$\Delta u_i|_{k=1} = \mu_i - \frac{\gamma}{2} \sigma_{ii} - \gamma \sum_{j \neq i} \sigma_{ij} x_j \tag{15}$$

TransformasiQUBO

$$E(\vec{x}) = -U(\vec{x}) \tag{16}$$

$$E(\vec{x}) = -\left(\frac{1}{k} \sum_{i=1}^{N} \mu_i x_i - \frac{\gamma}{2k^2} \sum_{i=1}^{N} \sum_{j=1}^{N} \sigma_{ij} x_i x_j \right) \tag{17}$$

$$E(\vec{x}) = \frac{\gamma}{2k^2} \sum_{i=1}^{N} \sum_{j=1}^{N} \sigma_{ij} x_i x_j - \frac{1}{k} \sum_{i=1}^{N} \mu_i x_i \tag{18}$$

$$\sum_{i=1}^{N} \sum_{j=1}^{N} \sigma_{ij} x_i x_j = \sum_{i=1}^{N} \sigma_{ii} x_i^2 + \sum_{i \neq j} \sigma_{ij} x_i x_j \tag{19}$$

$$x_i^2 = x_i, \quad x_i \in \{0, 1\} \tag{20}$$

$$E(\vec{x}) = \sum_{i=1}^{N} \left(\frac{\gamma \sigma_{ii}}{2k^2} - \frac{\mu_i}{k} \right) x_i + \frac{\gamma}{2k^2} \sum_{i \neq j} \sigma_{ij} x_i x_j \tag{21}$$

$$P(\vec{x}) = A\left(\sum_{i=1}^{N} x_i - k\right)^2 \tag{22}$$

$$P(\vec{x}) = A\left[\left(\sum_{i=1}^{N} x_i\right)^2 - 2k\sum_{i=1}^{N} x_i + k^2\right] \tag{23}$$

$$\left(\sum_{i=1}^{N} x_i\right)^2 = \sum_{i=1}^{N} x_i^2 + \sum_{i\neq j} x_i x_j = \sum_{i=1}^{N} x_i + \sum_{i\neq j} x_i x_j \tag{24}$$

$$P(\vec{x}) = \sum_{i=1}^{N} A(1-2k)x_i + \sum_{i\neq j} A x_i x_j + Ak^2 \tag{25}$$

$$H(\vec{x}) = E(\vec{x}) + P(\vec{x}) \tag{26}$$

$$H(\vec{x}) = \sum_{i=1}^{N}\left[\frac{\gamma\sigma_{ii}}{2k^2} - \frac{\mu_i}{k} + A(1-2k)\right] x_i + \sum_{i\neq j}\left(\frac{\gamma\sigma_{ij}}{2k^2} + A\right) x_i x_j + Ak^2 \tag{27}$$

$$H(\vec{x}) = \sum_{i=1}^{N} Q_{ii} x_i + \sum_{i\neq j} Q_{ij} x_i x_j + C \tag{28}$$

- $H(\vec{x})$: Hamiltonian total (QUBO).
- $Q_{ii} = \frac{\gamma\sigma_{ii}}{2k^2} - \frac{\mu_i}{k} + A(1-2k)$: Koefisien linear.
- $Q_{ij} = \frac{\gamma\sigma_{ij}}{2k^2} + A$: Koefisien kuadratik.
- $C = Ak^2$: Konstanta energi.

TransformasiIsing

$$x_i = \frac{s_i + 1}{2}, \quad s_i \in \{-1, 1\} \tag{29}$$

$$H(\vec{s}) = \sum_{i=1}^{N} Q_{ii}\left(\frac{s_i+1}{2}\right) + \sum_{i\neq j} Q_{ij}\left(\frac{s_i+1}{2}\right)\left(\frac{s_j+1}{2}\right) + C \tag{30}$$

$$\left(\frac{s_i+1}{2}\right)\left(\frac{s_j+1}{2}\right) = \frac{1}{4}(s_i s_j + s_i + s_j + 1) \tag{31}$$

$$H(\vec{s}) = \sum_{i=1}^{N} \frac{Q_{ii}}{2}(s_i + 1) + \sum_{i\neq j} \frac{Q_{ij}}{4}(s_i s_j + s_i + s_j + 1) + C \tag{32}$$

$$H(\vec{s}) = \sum_{i \neq j} \frac{Q_{ij}}{4} s_i s_j + \sum_{i=1}^{N} \left(\frac{Q_{ii}}{2} + \sum_{j \neq i} \frac{Q_{ij}}{2} \right) s_i + \left(\sum_{i=1}^{N} \frac{Q_{ii}}{2} + \sum_{i \neq j} \frac{Q_{ij}}{4} + C \right) \tag{33}$$

$$H(\vec{s}) = \sum_{i \neq j} J_{ij} s_i s_j + \sum_{i=1}^{N} h_i s_i + C_{Ising} \tag{34}$$

- $J_{ij} = \frac{Q_{ij}}{4}$: Kekuatan interaksi antar *spin*.
- $h_i = \frac{Q_{ii}}{2} + \sum_{j \neq i} \frac{Q_{ij}}{2}$: Medan magnet eksternal.
- $C_{Ising} = \sum_{i=1}^{N} \frac{Q_{ii}}{2} + \sum_{i \neq j} \frac{Q_{ij}}{4} + C$: Pergeseran energi konstan.

$$s_i \rightarrow \hat{Z}_i \tag{35}$$

$$\hat{Z}_i = \mathbb{I}_2 \otimes \mathbb{I}_2 \otimes \cdots \otimes \sigma_i^z \otimes \cdots \otimes \mathbb{I}_2 \tag{36}$$

$$\sigma^z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \mathbb{I}_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \tag{37}$$

$$\hat{H} = \sum_{i \neq j} J_{ij} \hat{Z}_i \hat{Z}_j + \sum_{i=1}^{N} h_i \hat{Z}_i + C_{Ising} \hat{\mathbb{I}} \tag{38}$$

$$E(\theta) = \langle \psi(\theta) | \hat{H} | \psi(\theta) \rangle \tag{39}$$

$$E(\theta) = \sum_{i \neq j} J_{ij} \langle \psi(\theta) | \hat{Z}_i \hat{Z}_j | \psi(\theta) \rangle + \sum_{i=1}^{N} h_i \langle \psi(\theta) | \hat{Z}_i | \psi(\theta) \rangle + C_{Ising} \tag{40}$$

Gradient Descent

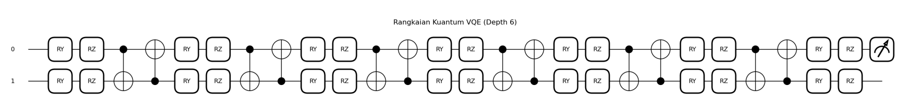

Figure 1: Representasi sirkuit kuantum variasional dengan struktur *ansatz* yang terdiri dari operator $\hat{U}_A$, $\hat{U}(\theta)$, dan $\hat{U}_B$ untuk optimasi parameter θ.

$$|\psi(\theta)\rangle = \hat{U}_B \hat{U}(\theta) \hat{U}_A |0\rangle \tag{41}$$

$$\hat{U}_A = \prod_{j=1}^{k} \hat{G}_j, \quad \hat{G}_j \in \mathbb{C}^{4\times 4} \tag{42}$$

$$\hat{U}(\theta) = \mathbb{I} \otimes \hat{R}_y(\theta) = \begin{pmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} & 0 & 0 \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} & 0 & 0 \\ 0 & 0 & \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ 0 & 0 & \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix} \tag{43}$$

$$\hat{U}_B = \prod_{j=k+1}^{m} \hat{G}_j, \quad \hat{G}_j \in \mathbb{C}^{4\times 4} \tag{44}$$

$$\hat{U}(\theta) = e^{-i\frac{\theta}{2}\hat{P}} \tag{45}$$

$$|\phi\rangle = \hat{U}_A|0\rangle \tag{46}$$

$$\hat{M} = \hat{U}_B^{\dagger}\hat{H}\hat{U}_B \tag{47}$$

$$E(\theta) = \langle\phi|\hat{U}^{\dagger}(\theta)\hat{M}\hat{U}(\theta)|\phi\rangle \tag{48}$$

$$\frac{\partial\hat{U}(\theta)}{\partial\theta} = -i\frac{1}{2}\hat{P}\hat{U}(\theta) \tag{49}$$

$$\frac{\partial\hat{U}^{\dagger}(\theta)}{\partial\theta} = i\frac{1}{2}\hat{U}^{\dagger}(\theta)\hat{P} \tag{50}$$

$$\frac{\partial E(\theta)}{\partial\theta} = \langle\phi|\frac{\partial\hat{U}^{\dagger}(\theta)}{\partial\theta}\hat{M}\hat{U}(\theta)|\phi\rangle + \langle\phi|\hat{U}^{\dagger}(\theta)\hat{M}\frac{\partial\hat{U}(\theta)}{\partial\theta}|\phi\rangle \tag{51}$$

$$\frac{\partial E(\theta)}{\partial\theta} = \frac{i}{2}\langle\phi|\hat{U}^{\dagger}(\theta)\hat{P}\hat{M}\hat{U}(\theta)|\phi\rangle - \frac{i}{2}\langle\phi|\hat{U}^{\dagger}(\theta)\hat{M}\hat{P}\hat{U}(\theta)|\phi\rangle \tag{52}$$

$$[\hat{P}, \hat{U}(\theta)] = 0 \implies \hat{P}\hat{U}(\theta) = \hat{U}(\theta)\hat{P} \tag{53}$$

$$\frac{\partial E(\theta)}{\partial\theta} = \frac{i}{2}\langle\phi|\hat{U}^{\dagger}(\theta)[\hat{P}, \hat{M}]\hat{U}(\theta)|\phi\rangle \tag{54}$$

$$|\psi(\theta)\rangle = \hat{U}(\theta)|\phi\rangle \tag{55}$$

$$\frac{\partial E(\theta)}{\partial\theta} = \frac{i}{2}\langle\psi(\theta)|[\hat{P}, \hat{M}]|\psi(\theta)\rangle \tag{56}$$

$$E(\theta + s) = \langle\psi(\theta)|e^{i\frac{s}{2}\hat{P}}\hat{M}e^{-i\frac{s}{2}\hat{P}}|\psi(\theta)\rangle \tag{57}$$

$$E(\theta + s) = \langle\psi(\theta)|\left[\cos^2\left(\frac{s}{2}\right)\hat{M} + i\sin\left(\frac{s}{2}\right)\cos\left(\frac{s}{2}\right)[\hat{P},\hat{M}] + \sin^2\left(\frac{s}{2}\right)\hat{P}\hat{M}\hat{P}\right]|\psi(\theta)\rangle \tag{58}$$

$$E(\theta - s) = \langle\psi(\theta)|e^{-i\frac{s}{2}\hat{P}}\hat{M}e^{i\frac{s}{2}\hat{P}}|\psi(\theta)\rangle \tag{59}$$

$$E(\theta - s) = \langle\psi(\theta)|\left[\cos^2\left(\frac{s}{2}\right)\hat{M} - i\sin\left(\frac{s}{2}\right)\cos\left(\frac{s}{2}\right)[\hat{P},\hat{M}] + \sin^2\left(\frac{s}{2}\right)\hat{P}\hat{M}\hat{P}\right]|\psi(\theta)\rangle \tag{60}$$

$$E(\theta + s) - E(\theta - s) = 2i\sin\left(\frac{s}{2}\right)\cos\left(\frac{s}{2}\right)\langle\psi(\theta)|[\hat{P},\hat{M}]|\psi(\theta)\rangle \tag{61}$$

$$E(\theta + s) - E(\theta - s) = i\sin(s)\langle\psi(\theta)|[\hat{P},\hat{M}]|\psi(\theta)\rangle \tag{62}$$

$$s = \frac{\pi}{2} \implies \sin\left(\frac{\pi}{2}\right) = 1 \tag{63}$$

$$E\left(\theta + \frac{\pi}{2}\right) - E\left(\theta - \frac{\pi}{2}\right) = i\langle\psi(\theta)|[\hat{P},\hat{M}]|\psi(\theta)\rangle \tag{64}$$

$$\frac{\partial E(\theta)}{\partial\theta} = \frac{1}{2}\left(i\langle\psi(\theta)|[\hat{P},\hat{M}]|\psi(\theta)\rangle\right) \tag{65}$$

$$\frac{\partial E(\theta)}{\partial\theta} = \frac{1}{2}\left[E\left(\theta + \frac{\pi}{2}\right) - E\left(\theta - \frac{\pi}{2}\right)\right] \tag{66}$$

$$\theta_{n+1} = \theta_n - \eta\frac{\partial E(\theta_n)}{\partial\theta} \tag{67}$$

$$\theta^{(0)} = \frac{\pi}{4}, \quad s = \frac{\pi}{2} \tag{68}$$

$$\theta_+ = \theta^{(0)} + s = \frac{3\pi}{4} \tag{69}$$

$$P(1)_+ = \sin^2\left(\frac{3\pi}{8}\right) \approx 0.8535 \tag{70}$$

$$E(\theta_+) = 10(P(0)_+ - P(1)_+) = 10(0.1465 - 0.8535) = -7.07 \tag{71}$$

$$\theta_- = \theta^{(0)} - s = -\frac{\pi}{4} \tag{72}$$

$$P(1)_- = \sin^2\left(-\frac{\pi}{8}\right) \approx 0.1465 \tag{73}$$

$$E(\theta_-) = 10(P(0)_- - P(1)_-) = 10(0.8535 - 0.1465) = 7.07 \tag{74}$$

$$\frac{\partial E(\theta^{(0)})}{\partial \theta} = \frac{1}{2}(E(\theta_+) - E(\theta_-)) \tag{75}$$

$$\frac{\partial E(\theta^{(0)})}{\partial \theta} = \frac{1}{2}(-7.07 - 7.07) = -7.07 \tag{76}$$

$$\theta^{(1)} = \theta^{(0)} - \eta \frac{\partial E(\theta^{(0)})}{\partial \theta} = \frac{\pi}{4} + 7.07\eta \tag{77}$$

SPSA

$$X \sim \text{Bernoulli}(p = 0.5) \implies P(X = 1) = 0.5, \quad P(X = 0) = 0.5 \tag{78}$$

- X : variabel acak Bernoulli

$$\Delta_i = 2X - 1 \implies \Delta_i \in \{-1, 1\} \tag{79}$$

- Δ_i : variabel acak Rademacher komponen i

$$\vec{\Delta}_k = (\Delta_{k1}, \Delta_{k2}, \dots, \Delta_{kp})^T \tag{80}$$

- $\vec{\Delta}_k$: vektor perturbasi stokastik pada iterasi k

$$E_+ = E(\vec{\theta}_k + c_k\vec{\Delta}_k) \tag{81}$$

$$E_- = E(\vec{\theta}_k - c_k\vec{\Delta}_k) \tag{82}$$

- $\vec{\theta}_k$: vektor parameter pada iterasi k
- c_k : *perturbation step* pada iterasi k

$$E_+ = E(\vec{\theta}_k) + c_k\vec{\Delta}_k^T \nabla E(\vec{\theta}_k) + \frac{1}{2}c_k^2\vec{\Delta}_k^T \nabla^2 E(\vec{\theta}_k)\vec{\Delta}_k + O(c_k^3) \tag{83}$$

$$E_- = E(\vec{\theta}_k) - c_k\vec{\Delta}_k^T \nabla E(\vec{\theta}_k) + \frac{1}{2}c_k^2\vec{\Delta}_k^T \nabla^2 E(\vec{\theta}_k)\vec{\Delta}_k - O(c_k^3) \tag{84}$$

$$E_+ - E_- = 2c_k\vec{\Delta}_k^T \nabla E(\vec{\theta}_k) + O(c_k^3) \tag{85}$$

$$E_+ - E_- = 2c_k \sum_{j=1}^{p} \Delta_{kj} \frac{\partial E(\vec{\theta}_k)}{\partial \theta_j} + O(c_k^3) \tag{86}$$

$$\hat{g}_{ki} = \frac{E_+ - E_-}{2c_k\Delta_{ki}} \tag{87}$$

- $\hat{g}_{ki}$: estimator gradien komponen i pada iterasi k

$$\hat{g}_{ki} = \frac{2c_k\Delta_{ki}\frac{\partial E}{\partial \theta_i} + 2c_k\sum_{j\neq i}^{p}\Delta_{kj}\frac{\partial E}{\partial \theta_j} + O(c_k^3)}{2c_k\Delta_{ki}} \tag{88}$$

$$\hat{g}_{ki} = \frac{\partial E(\vec{\theta}_k)}{\partial \theta_i} + \sum_{j\neq i}^{p}\frac{\Delta_{kj}}{\Delta_{ki}}\frac{\partial E(\vec{\theta}_k)}{\partial \theta_j} + O(c_k^2) \tag{89}$$

$$\mathbb{E}[\hat{g}_{ki}|\vec{\theta}_k] = \mathbb{E}\left[\frac{\partial E(\vec{\theta}_k)}{\partial \theta_i} + \sum_{j\neq i}^{p}\frac{\Delta_{kj}}{\Delta_{ki}}\frac{\partial E(\vec{\theta}_k)}{\partial \theta_j} + O(c_k^2)\right] \tag{90}$$

$$\mathbb{E}\left[\frac{\Delta_{kj}}{\Delta_{ki}}\right] = \mathbb{E}[\Delta_{kj}]\mathbb{E}\left[\frac{1}{\Delta_{ki}}\right] \tag{91}$$

$$\mathbb{E}[\Delta_{kj}] = 0.5(1) + 0.5(-1) = 0 \tag{92}$$

$$\mathbb{E}\left[\frac{\Delta_{kj}}{\Delta_{ki}}\right] = 0 \cdot 0 = 0, \quad \forall j \neq i \tag{93}$$

$$\mathbb{E}[\hat{g}_{ki}|\vec{\theta}_k] = \frac{\partial E(\vec{\theta}_k)}{\partial \theta_i} + O(c_k^2) \tag{94}$$

$$\hat{H} = 10\hat{Z}_1 + 5\hat{Z}_2 \tag{95}$$

$$\hat{H} = 10(\hat{Z}\otimes\hat{I}) + 5(\hat{I}\otimes\hat{Z}) \tag{96}$$

Konfigurasi Awal:

- Hiperparameter: $a = 0.1, c = 0.1, \alpha = 0.602, \gamma = 0.101$
- Parameter Awal: $\vec{\theta}^{(0)} = [0.7854, 0.7854]^T$
- Fungsi Energi: $E(\vec{\theta}) = 10\cos(\theta_1) + 5\cos(\theta_2)$

Iterasi 1 ($k = 1$)

$$a_1 = \frac{0.1}{1^{0.602}} = 0.1, \quad c_1 = \frac{0.1}{1^{0.101}} = 0.1 \tag{97}$$

$$\vec{\Delta}_1 = [1, -1]^T \tag{98}$$

$$\vec{\theta}^{+} = \vec{\theta}^{(0)} + c_1\vec{\Delta}_1 = [0.8854, 0.6854]^T \tag{99}$$

$$\vec{\theta}^{-} = \vec{\theta}^{(0)} - c_1\vec{\Delta}_1 = [0.6854, 0.8854]^T \tag{100}$$

$$E(\vec{\theta}^{+}) = 10\cos(0.8854) + 5\cos(0.6854) = 10.1995 \tag{101}$$

$$E(\vec{\theta}^{-}) = 10\cos(0.6854) + 5\cos(0.8854) = 10.9055 \tag{102}$$

$$\hat{g}_{11} = \frac{10.1995 - 10.9055}{2(0.1)(1)} = -3.53 \tag{103}$$

$$\hat{g}_{12} = \frac{10.1995 - 10.9055}{2(0.1)(-1)} = 3.53 \tag{104}$$

$$\vec{\theta}^{(1)} = \vec{\theta}^{(0)} - a_1\hat{g}_1 = [1.1384, 0.4324]^T \tag{105}$$

$$E(\vec{\theta}^{(1)}) = 10\cos(1.1384) + 5\cos(0.4324) = 8.7306 \tag{106}$$

Iterasi 2 $(k = 2)$

$$a_2 = \frac{0.1}{2^{0.602}} = 0.0659, \quad c_2 = \frac{0.1}{2^{0.101}} = 0.0932 \tag{107}$$

$$\vec{\Delta}_2 = [-1, -1]^T \tag{108}$$

$$\vec{\theta}^{+} = \vec{\theta}^{(1)} + c_2\vec{\Delta}_2 = [1.0451, 0.3392]^T \tag{109}$$

$$\vec{\theta}^{-} = \vec{\theta}^{(1)} - c_2\vec{\Delta}_2 = [1.2316, 0.5257]^T \tag{110}$$

$$E(\vec{\theta}^{+}) = 10\cos(1.0451) + 5\cos(0.3392) = 9.7331 \tag{111}$$

$$E(\vec{\theta}^{-}) = 10\cos(1.2316) + 5\cos(0.5257) = 7.6522 \tag{112}$$

$$\hat{g}_{21} = \frac{9.7331 - 7.6522}{2(0.0932)(-1)} = -11.1587 \tag{113}$$

$$\hat{g}_{22} = \frac{9.7331 - 7.6522}{2(0.0932)(-1)} = -11.1587 \tag{114}$$

$$\vec{\theta}^{(2)} = \vec{\theta}^{(1)} - a_2\hat{g}_2 = [1.8735, 1.1676]^T \tag{115}$$

$$E(\vec{\theta}^{(2)}) = 10\cos(1.8735) + 5\cos(1.1676) = -1.0197 \tag{116}$$

Catatan: Karena $\Delta_{21} = \Delta_{22} = -1$, maka $\hat{g}_{21} = \hat{g}_{22}$. Estimator gradien SPSA tidak membedakan kontribusi parameter individual saat perturbasi seragam, berbeda dengan gradien analitik: $\nabla E = [-9.0795, -2.0954]^T$.

Ringkasan Konvergensi:

Iterasi (k)	θ_1	θ_2	$E(\vec{\theta})$	Jarak ke -15
0	0.7854	0.7854	10.6066	25.6066
1	1.1384	0.4324	8.7306	23.7306
2	1.8735	1.1676	-1.0197	13.9803

Quantum Computing

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \tag{117}$$

- $|0\rangle, |1\rangle$: Vektor basis komputasi kuantum (*computational basis states*).

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{118}$$

- $\sigma_x, \sigma_y, \sigma_z$: Matriks Pauli X, Y, dan Z.

$$\hat{R}_y(\theta) = \exp\left(-i\frac{\theta}{2}\sigma_y\right) \tag{119}$$

$$\hat{R}_y(\theta) = \hat{I}\cos\left(\frac{\theta}{2}\right) - i\sigma_y \sin\left(\frac{\theta}{2}\right) \tag{120}$$

$$\hat{R}_y(\theta) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \cos\left(\frac{\theta}{2}\right) - i \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \sin\left(\frac{\theta}{2}\right) \tag{121}$$

$$\hat{R}_y(\theta) = \begin{pmatrix} \cos t(\frac{\theta}{2} & -\sin\left(\frac{\theta}{2}\right) \\ \sin\left(\frac{\theta}{2}\right) & \cos\left(\frac{\theta}{2}\right) \end{pmatrix} \tag{122}$$

$$\hat{R}_z(\theta) = \exp\left(-i\frac{\theta}{2}\sigma_z\right) \tag{123}$$

$$\hat{R}_z(\theta) = \hat{I}\cos\left(\frac{\theta}{2}\right) - i\sigma_z \sin\left(\frac{\theta}{2}\right) \tag{124}$$

$$\hat{R}_z(\theta) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \cos\left(\frac{\theta}{2}\right) - i \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \sin\left(\frac{\theta}{2}\right) \tag{125}$$

$$\hat{R}_z(\theta) = \begin{pmatrix} \cos\left(\frac{\theta}{2}\right) - i\sin\left(\frac{\theta}{2}\right) & 0 \\ 0 & \cos\left(\frac{\theta}{2}\right) + i\sin\left(\frac{\theta}{2}\right) \end{pmatrix} \tag{126}$$

$$\hat{R}_z(\theta) = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix} \tag{127}$$

$$\text{CNOT} = |0\rangle\langle 0| \otimes \hat{I} + |1\rangle\langle 1| \otimes \sigma_x \tag{128}$$

$$\text{CNOT} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \otimes \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \otimes \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \tag{129}$$

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \tag{130}$$

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \tag{131}$$

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \tag{132}$$

- α, β : Amplitudo probabilitas ($\alpha, \beta \in \mathbb{C}$).

$$\langle\psi|\psi\rangle = 1 \tag{133}$$

$$(\alpha^*\langle 0| + \beta^*\langle 1|)(\alpha|0\rangle + \beta|1\rangle) = 1 \tag{134}$$

$$|\alpha|^2\langle 0|0\rangle + \alpha^*\beta\langle 0|1\rangle + \beta^*\alpha\langle 1|0\rangle + |\beta|^2\langle 1|1\rangle = 1 \tag{135}$$

$$|\alpha|^2 + |\beta|^2 = 1 \tag{136}$$

$$\hat{H}|\psi_n\rangle = E_n|\psi_n\rangle \tag{137}$$

- $\hat{H}$: Operator Hamiltonian.
- $|\psi_n\rangle$: Eigenstate ke-n.
- E_n : Eigenvalue energi ke-n.

$$E(\theta) = \frac{\langle\psi(\theta)|\hat{H}|\psi(\theta)\rangle}{\langle\psi(\theta)|\psi(\theta)\rangle} \tag{138}$$

$$E(\theta) = \langle\psi(\theta)|\hat{H}|\psi(\theta)\rangle \tag{139}$$

- $E(\theta)$: Nilai ekspektasi energi (*cost function*).
- $|\psi(\theta)\rangle$: *Trial ansatz* terparameterisasi.

$$E(\theta) = \langle\psi(\theta)| \left(\sum_n E_n|\psi_n\rangle\langle\psi_n| \right) |\psi(\theta)\rangle \tag{140}$$

$$E(\theta) = \sum_n E_n|\langle\psi_n|\psi(\theta)\rangle|^2 \geq E_0 \tag{141}$$

- Prinsip Variasional Rayleigh-Ritz: $E(\theta) \geq E_0$.